SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1711**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I **T. Niranjan Kumar (192473007)** (Name / Reg No) certify that this submission is my original work and that I have adhered*
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Niranjan

SELECTED RESEARCH ARTICLE

Joshi, R. D., & Dhakal, C. K. (2021). Predicting Type 2 Diabetes Using Logistic Regression and Machine Learning Approaches. International Journal of Environmental Research and Public Health, 18(14), 7346. DOI: 10.3390/ijerph18147346.

Domain/Application Area: Healthcare – Type 2 diabetes risk prediction.

Specific ML Problem: Predicting whether a patient is likely to have Type 2 diabetes and identifying important risk factors using Logistic Regression and a Decision Tree classification approach.

DOI Link: https://doi.org/10.3390/ijerph18147346

TASK 1 – ARTICLE SUMMARY

(a) Full Citation, Domain and ML Problem

The selected article is a 2021 peer-reviewed research article published in the International Journal of Environmental Research and Public Health. It belongs to the healthcare analytics domain and studies Type 2 diabetes prediction. The authors use Logistic Regression to study risk factors and a Decision Tree to complement and validate the classification analysis. The work focuses on Pima Indian women and examines how clinical and demographic variables can be used to predict diabetes risk.

(b) Article Summary (150–200 words)

The main objective of the article is to predict Type 2 diabetes and identify the most important factors associated with the disease by combining a statistical learning method with a machine learning classification method. The authors use Logistic Regression to estimate the relationship between patient characteristics and diabetes outcome, and then use a Decision Tree to complement the analysis through rule-based classification. In my understanding, the research gap addressed by the study is that diabetes prediction should not depend only on a black-box prediction result; healthcare users also need to understand which variables are most strongly associated with risk. The authors attempt to fill this gap by combining interpretable statistical analysis with a classification tree. Their results identify glucose, pregnancy, BMI, diabetes pedigree function, and age as important predictors in the preferred analysis. The classification tree further highlights glucose, BMI, and age as important variables, while a larger tree also includes pregnancy and diabetes pedigree function. The preferred specification achieved 78.26% prediction accuracy with a 21.74% cross-validation error rate. Overall, the article shows how statistical learning and decision-tree learning can support early diabetes risk assessment while also providing understandable information about major risk factors.

TASK 2 – METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

Technique 1 – Logistic Regression: The article uses Logistic Regression to estimate the relationship between predictor variables and the binary diabetes outcome. This is a statistical learning approach because it models the probability of a class outcome from observed features.

Technique 2 – Decision Tree: The authors also construct classification trees. The tree recursively selects informative variables and produces understandable decision rules. The reported analyses include a six-fold classification tree and a ten-node tree.

Dataset: The study uses data on Pima Indian women. The variables discussed include glucose, pregnancy, BMI, diabetes pedigree function, and age, with diabetes outcome as the target. The available article information used for this review supports these features and the reported model results. Detailed preprocessing counts and every dataset-preparation operation are not fully specified in the assessment file, so this review does not invent unsupported preparation steps.

(b) Mapping to CO4, Strength and Limitation

CO4 Mapping

- Inductive Learning: Both models learn general patterns from historical patient examples and apply them to predict diabetes outcomes.
- Decision Trees: The article explicitly uses classification trees to discover interpretable feature-based decision rules.
- Statistical Learning: Logistic Regression is used to estimate diabetes risk and identify significant predictors.
- Reinforcement Learning: This specific article does not use Reinforcement Learning; therefore, RL is not a direct component of the selected study.

Methodological Strength

A major strength is the combination of Logistic Regression and Decision Trees. Logistic Regression provides a statistical view of important predictors, while the Decision Tree offers interpretable classification rules. Using both methods makes the analysis more understandable than relying on only one model.

Methodological Limitation

A limitation is the use of a specific population of Pima Indian women, which may restrict how directly the findings generalise to other populations. In addition, the reported accuracy alone is not sufficient for a complete medical-risk evaluation because class-specific recall, precision, calibration and external validation are also important.

TASK 3 – RESULTS & CRITICAL EVALUATION

(a) Key Results

The authors report that the preferred specification achieved 78.26% prediction accuracy and a 21.74% cross-validation error rate. The study identifies five main predictors in its preferred Logistic Regression analysis: glucose, pregnancy, BMI, diabetes pedigree function, and age. The classification tree analyses highlight glucose, BMI and age as important in the six-fold tree, while the ten-node tree includes glucose, BMI, pregnancy, diabetes pedigree function and age.

Model / Analysis	Key Finding	Reported Result
Logistic Regression	Five main predictors identified	Glucose, pregnancy, BMI, diabetes pedigree function, age
Six-fold Classification Tree	Important predictors highlighted	Glucose, BMI, age
Ten-node Classification Tree	Broader predictor set	Glucose, BMI, pregnancy, diabetes pedigree function, age
Preferred Specification	Predictive performance	Accuracy: 78.26%; CV error: 21.74%

(b) Critical Evaluation

The reported 78.26% accuracy appears plausible for a diabetes prediction problem, but accuracy should not be interpreted as the only indicator of clinical usefulness. A dataset with unequal class frequencies can produce a reasonable accuracy even when the model performs poorly for the diabetic class. Therefore, additional metrics such as recall, precision, F1-score, AUC-ROC and calibration would strengthen the evaluation.

Potential bias can arise from the study population because the data concern Pima Indian women, so performance may differ for men or for other ethnic and geographic populations. Evaluation bias can also arise if model selection and testing are not separated carefully. Additional experiments that would strengthen the findings include external validation on another population, repeated cross-validation, comparison with additional modern classifiers, class-imbalance analysis, calibration testing and subgroup performance analysis.

(c) Real-World Applicability and Scaling Challenges

The technique could support diabetes-risk screening tools in primary healthcare, preventive-care programs and clinical decision-support systems. It may also help identify people who need further medical testing. However, industry-scale deployment faces several challenges: obtaining high-quality and representative patient data, missing or inconsistent records, data privacy and security, integration with hospital information systems, model monitoring, fairness across patient groups and the need for clinician oversight. The system should support medical professionals rather than replace diagnosis by qualified clinicians.

TASK 4 – REFLECTION & LEARNING OUTCOME

(a) Three Key Insights Connected to CO4

Insight 1 – Statistical Learning: Logistic Regression is not only a prediction model; it also helps analyse which variables are associated with a binary outcome. This deepened my understanding of the CO4 topic of statistical learning methods.

Insight 2 – Decision Trees: A Decision Tree can transform patient features into understandable classification rules. This connects directly to the CO4 topic of decision-tree algorithms and shows why interpretability is valuable in healthcare.

Insight 3 – Inductive Learning and Model Evaluation: The models learn general patterns from observed examples and then apply them to new patients. This demonstrates inductive learning. The article also shows that model performance must be evaluated with validation methods rather than trusting training performance alone.

Additional Learning Point: The selected article does not use Reinforcement Learning, which also helped me distinguish supervised prediction problems from sequential decision-making problems. RL would be more suitable when an agent repeatedly chooses actions and receives rewards over time.

(b) Proposed Extension

If I were to extend this research, I would test the combined Logistic Regression and Decision Tree approach on a larger, multi-hospital dataset containing diverse patients and additional lifestyle variables. I would use stratified cross-validation, external validation and fairness analysis across relevant patient groups. I would also compare the original models with calibrated ensemble methods while keeping at least one interpretable baseline model. This extension is feasible because the models are computationally practical for structured healthcare data. The expected impact would be stronger evidence of generalisability, better understanding of subgroup performance and a more reliable pathway toward real-world clinical decision support.

FINAL CONCLUSION

This article review demonstrates the application of CO4 concepts through Logistic Regression, Decision Tree classification and inductive learning for Type 2 diabetes prediction. The selected study provides an interpretable example of combining statistical learning with machine learning. Its main contribution is identifying important diabetes predictors while producing a practical classification result. At the same time, broader validation and richer evaluation metrics would be necessary before using the approach at large clinical scale.